Heart of Algebra

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solving linear equations**

Solve for x: $4x - 9 = 23$.

$4x = 32$, so $x = 8$.

2 **[Foundational]** • **Target: Solving linear equations**

Solve for x: $\frac{3x + 2}{5} = 4$.

Multiply both sides by 5: $3x + 2 = 20$, so $3x = 18$, giving $x = 6$.

3 **[Foundational]** • **Target: Solving linear equations**

Solve for x: $6(x - 3) = 2(x + 9)$.

Expand: $6x - 18 = 2x + 18$. Subtract $2x$: $4x - 18 = 18$, so $4x = 36$, giving $x = 9$.

4 **[Foundational]** • **Target: Linear inequalities**

Solve $5x - 3 > 12$.

$5x > 15$, so $x > 3$.

5 **[Foundational]** • **Target: Linear inequalities**

Solve $-2x + 7 \leq 15$.

Subtract 7: $-2x \leq 8$. Divide by -2 and flip the inequality: $x \geq -4$.

6 **[Foundational]** • **Target: Linear inequalities**

Solve $3(x - 2) \geq x + 4$.

Expand: $3x - 6 \geq x + 4$. Subtract x: $2x - 6 \geq 4$, so $2x \geq 10$, giving $x \geq 5$.

7 **[Foundational]** • **Target: Slope and equations of lines**

Find the slope of the line through $(2, 5)$ and $(6, 13)$.

$m = \frac{13 - 5}{6 - 2} = \frac{8}{4} = 2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Slope and equations of lines

Write the equation, in slope-intercept form, of the line with slope -3 passing through $(0, 7)$.

$y = -3x + 7$.

9 [Intermediate] • Target: Slope and equations of lines

Find the slope of a line perpendicular to $y = \frac{2}{5}x - 1$.

A perpendicular slope is the negative reciprocal of $\frac{2}{5}$, which is $-\frac{5}{2}$.

10 [Intermediate] • Target: Systems of equations by substitution

Solve the system: $y = 3x - 4$ and $2x + y = 11$.

Substitute: $2x + (3x - 4) = 11$, so $5x - 4 = 11$, giving $5x = 15$, $x = 3$. Then $y = 3(3) - 4 = 5$. Solution: $(3, 5)$.

11 [Intermediate] • Target: Systems of equations by substitution

Solve the system: $x - 2y = 1$ and $3x + y = 17$.

From the first equation, $x = 1 + 2y$. Substitute: $3(1 + 2y) + y = 17$, so $3 + 6y + y = 17$, giving $7y = 14$, $y = 2$. Then $x = 1 + 2(2) = 5$. Solution: $(5, 2)$.

12 [Intermediate] • Target: Systems of equations by elimination

Solve the system: $3x + 2y = 19$ and $x - 2y = 1$, using elimination.

Adding the two equations eliminates y: $4x = 20$, so $x = 5$. Substituting into $x - 2y = 1$: $5 - 2y = 1$, so $2y = 4$, giving $y = 2$. Solution: $(5, 2)$.

13 [Intermediate] • Target: Systems of equations by elimination

Solve the system: $4x - y = 10$ and $2x + y = 14$, using elimination.

Adding the two equations eliminates y: $6x = 24$, so $x = 4$. Substituting into $2x + y = 14$: $8 + y = 14$, so $y = 6$. Solution: $(4, 6)$.

14 [Intermediate] • Target: Absolute value equations

Solve $|2x - 5| = 9$.

$2x - 5 = 9 \Rightarrow x = 7$, or $2x - 5 = -9 \Rightarrow x = -2$. Solutions: $x = 7$ or $x = -2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Absolute value equations • Cross-Lesson: 2.01 Quadratics & Polynomials

Solve $|x + 4| = 10$.

$x + 4 = 10 \Rightarrow x = 6$, or $x + 4 = -10 \Rightarrow x = -14$. Solutions: $x = 6$ or $x = -14$.

16 [Challenging] • Target: Interpreting linear models • Cross-Lesson: 3.01 Functions & Graphs

A phone plan costs $C = 20 + 0.05m$ dollars for m minutes of calls. Interpret the meaning of the numbers 20 and 0.05 in this context.

The 20 represents a fixed monthly fee (the cost when $m = 0$ minutes are used), and the 0.05 represents the cost per minute — each additional minute of calls adds 0.05 dollars to the bill.

17 [Challenging] • Target: Interpreting linear models • Cross-Lesson: 4.01 Ratios, Rates & Data

Company A charges $C = 30 + 0.20d$ and Company B charges $C = 0.35d$ for a d-mile trip. Find the mileage at which both companies charge the same amount, and state which company is cheaper for a 150-mile trip.

Set equal: $30 + 0.2d = 0.35d$, so $30 = 0.15d$, giving $d = 200$ miles. At 150 miles: Company A charges $30 + 0.2(150) = 60$ dollars; Company B charges $0.35(150) = 52.5$ dollars. Company B is cheaper at 150 miles.

18 [Challenging] • Target: Systems with no solution or infinitely many solutions • Cross-Lesson: 5.01 Statistics & Probability

For what value of k does the equation $6x + 4 = kx + 4$ have infinitely many solutions?

For infinitely many solutions the equation must hold for every x, so the coefficients of x must match: $k = 6$ (the constants already agree, $4 = 4$).

19 [Challenging] • Target: Systems with no solution or infinitely many solutions • Cross-Lesson: 6.01 Geometry

Explain why the system $2x - 3y = 6$ and $4x - 6y = 10$ has no solution.

Multiplying the first equation by 2 gives $4x - 6y = 12$. This contradicts the second equation, $4x - 6y = 10$ — the same left-hand expression cannot equal two different values, so the lines are parallel and never intersect.

20 [Challenging] • Target: Compound inequalities • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Solve the compound inequality $-3 < 2x + 1 \leq 9$.

Subtract 1 from all three parts: $-4 < 2x \leq 8$. Divide by 2: $-2 < x \leq 4$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Solving linear equations

Solve for x: $-2x + 19 = 23$.

a. -4

b. 0

c. -2 ✓

d. 2

$-2x = 4$, so $x = -2$.

22 [Foundational] • Target: Linear inequalities

Solve for x: $2x + 11 = 13$.

a. -1

b. 3

c. 4

d. 1 ✓

$2x = 2$, so $x = 1$.

23 [Foundational] • Target: Slope and equations of lines

Solve for x: $4x - 14 = 2$.

a. 6

b. -4

c. 4 ✓

d. 2

$4x = 16$, so $x = 4$.

24 [Foundational] • Target: Systems of equations by substitution

Solve for x: $6x - 2 = 16$.

a. 5

b. 3 ✓

c. -3

d. 1

$6x = 18$, so $x = 3$.

25 [Foundational] • Target: Systems of equations by elimination

Solve for x: $-8x + 1 = 9$.

a. 2

b. -3

c. 1

d. -1 ✓

$-8x = 8$, so $x = -1$.

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26 [Foundational] • Target: Absolute value equations

Solve for x: $8x - 4 = -124$.

a. -17

b. 15

c. -15 ✓

d. -13

$8x = -120$, so $x = -15$.

27 [Foundational] • Target: Interpreting linear models

Solve for x: $4x - 11 = -15$.

a. 2

b. 1

c. -1 ✓

d. -3

$4x = -4$, so $x = -1$.

28 [Foundational] • Target: Systems with no solution or infinitely many solutions

Solve the system: $4x + y = -14$ and $4x - y = 6$, using elimination.

a. $(-1, -12)$

b. $(-1, -10)$ ✓

c. $(1, -10)$

d. $(-10, -1)$

Adding the two equations eliminates y: $8x = -8$, so $x = -1$. Substituting into the first equation: $4(-1) + y = -14$, so $y = -10$. Solution: $(-1, -10)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Compound inequalities

Solve the system: $6x + 4y = -44$ and $6x - 4y = 20$, using elimination.

a. $(-2, -10)$

b. $(-8, -2)$

c. $(-2, -8)$ ✓

d. $(0, -8)$

Adding the two equations eliminates y: $12x = -24$, so $x = -2$. Substituting into the first equation: $6(-2) + 4y = -44$, so $4y = -32$, giving $y = -8$. Solution: $(-2, -8)$.

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30 [Intermediate] • Target: Multi-step word problems

Solve the system: $2x + 4y = -6$ and $2x - 4y = 18$, using elimination.

a. $(-3, 3)$

b. $(5, -3)$

c. $(3, -5)$

d. $(3, -3)$ ✓

Adding the two equations eliminates y: $4x = 12$, so $x = 3$. Substituting into the first equation: $2(3) + 4y = -6$, so $4y = -12$, giving $y = -3$. Solution: $(3, -3)$.

31 [Intermediate] • Target: A well-designed word problem: ticket sales at a community theater

Solve the system: $2x + 4y = 40$ and $2x - 4y = -24$, using elimination.

a. $(8, 4)$

b. $(4, 6)$

c. $(4, 8)$ ✓

d. $(6, 8)$

Adding the two equations eliminates y: $4x = 16$, so $x = 4$. Substituting into the first equation: $2(4) + 4y = 40$, so $4y = 32$, giving $y = 8$. Solution: $(4, 8)$.

32 [Intermediate] • Target: Multi-Step Linear Equations & Identity Conditions

Solve the system: $3x + y = -17$ and $3x - y = -37$, using elimination.

a. $(10, -9)$

b. $(-7, 10)$

c. $(-9, 10)$ ✓

d. $(-9, 8)$

Adding the two equations eliminates y: $6x = -54$, so $x = -9$. Substituting into the first equation: $3(-9) + y = -17$, so $y = 10$. Solution: $(-9, 10)$.

33 [Intermediate] • Target: Linear Inequalities & Feasible Regions

Solve the system: $x - 5y = -48$ and $x + 5y = 52$, using elimination.

a. $(2, 8)$

b. $(4, 10)$

c. $(2, 10)$ ✓

d. $(10, 2)$

Adding the two equations eliminates y: $2x = 4$, so $x = 2$. Substituting into the first equation: $1(2) - 5y = -48$, so $-5y = -50$, giving $y = 10$. Solution: $(2, 10)$.

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34 [Intermediate] • Target: Systems of Linear Equations

Solve the system: $5x - y = -24$ and $5x + y = -6$, using elimination.

a. $(-1, 9)$

b. $(-3, 7)$

c. $(-3, 9)$ ✓

d. $(9, -3)$

Adding the two equations eliminates y: $10x = -30$, so $x = -3$. Substituting into the first equation: $5(-3) - y = -24$, so $-y = -9$, giving $y = 9$. Solution: $(-3, 9)$.

35 [Intermediate] • Target: Rate, Unit Conversions & Real-World Modeling

Solve $|2x - 12| = 26$.

a. 19

b. $-7, 19$ ✓

c. $7, -19$

d. $-7, 20$

$2x - 12 = 26 \Rightarrow x = 19$, or $2x - 12 = -26 \Rightarrow x = -7$. Solutions: $x = 19$ or $x = -7$.

36 [Intermediate] • Target: Solving linear equations

Solve $|2x - 7| = 21$.

a. 14

b. $7, -14$

c. $-7, 14$ ✓

d. $-7, 15$

$2x - 7 = 21 \Rightarrow x = 14$, or $2x - 7 = -21 \Rightarrow x = -7$. Solutions: $x = 14$ or $x = -7$.

37 [Intermediate] • Target: Linear inequalities

Solve $-4x - 2 \geq -42$.

a. $x \geq 10$

b. $x \leq 10$ ✓

c. $x \leq 12$

d. $x \leq 8$

Subtract -2: $-4x \geq -40$. Divide by -4 and flip the inequality: $x \leq 10$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Slope and equations of lines • Cross-Lesson: 5.01 Statistics & Probability

Solve $-7x - 3 \geq 53$.

a. $x \leq -8$ ✓

b. $x \leq -10$

c. $x \leq -6$

d. $x \geq -8$

Subtract -3: $-7x \geq 56$. Divide by -7 and flip the inequality: $x \leq -8$.

39 [Challenging] • Target: Systems of equations by substitution • Cross-Lesson: 6.01 Geometry

Two numbers have a sum of 114 and a difference of 16. Find the larger number.

a. 63

b. 49

c. 67

d. 65 ✓

Let the numbers be x and y with $x + y = 114$ and $x - y = 16$. Adding: $2x = 130$, so $x = 65$. Then $y = 114 - 65 = 49$. The larger number is 65.

40 [Challenging] • Target: Systems of equations by elimination • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Two numbers have a sum of 147 and a difference of 15. Find the larger number.

a. 81 ✓

b. 79

c. 66

d. 83

Let the numbers be x and y with $x + y = 147$ and $x - y = 15$. Adding: $2x = 162$, so $x = 81$. Then $y = 147 - 81 = 66$. The larger number is 81.

41 [Challenging] • Target: Multi-Step Linear Equations & Identity Conditions • Cross-Lesson: 8.01 Exponentials & Logarithms

If $5(2x - 1) - 3(x + 4) = 4(x - 3) + 7$, what is the value of x?

a. 4

b. 6

c. 8

d. 12 ✓

Expand both sides: $10x - 5 - 3x - 12 = 4x - 12 + 7 \Rightarrow 7x - 17 = 4x - 5$. Subtract $4x$ and add 17: $3x = 12 \Rightarrow x = 4$.

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42 [Challenging] • Target: Multi-Step Linear Equations & Identity Conditions • Cross-Lesson: 2.01 Quadratics & Polynomials

If $\frac{2x-5}{3} = \frac{x+4}{2}$, what is the value of $3x + 1$?

a. 22

b. 64

c. 67 ✓

d. 70

Cross multiply: $2(2x - 5) = 3(x + 4) \Rightarrow 4x - 10 = 3x + 12 \Rightarrow x = 22$. Then $3x + 1 = 3(22) + 1 = 67$.

43 [Challenging] • Target: Linear Inequalities & Feasible Regions • Cross-Lesson: 3.01 Functions & Graphs

If $-3x + 7 \geq 22$, what is the greatest possible value of x?

a. -5 ✓

b. -3

c. 5

d. 15

Subtract 7: $-3x \geq 15$. Divide by -3 (reverse inequality sign): $x \leq -5$. The maximum value is -5.

44 [Challenging] • Target: Systems of Linear Equations • Cross-Lesson: 4.01 Ratios, Rates & Data

For what value of k does the system $kx + 4y = 8$ and $3x + 2y = 5$ have no solution?

a. 3

b. 6 ✓

c. -6

d. 12

Parallel lines with no solution require equal slopes: $\frac{k}{3} = \frac{4}{2} = 2 \Rightarrow k = 6$.

45 [Challenging] • Target: Rate, Unit Conversions & Real-World Modeling • Cross-Lesson: 5.01 Statistics & Probability

A car uses 8 liters of fuel per 100 kilometers. How many liters of fuel are required for a trip of 350 kilometers?

a. 24

b. 26

c. 28 ✓

d. 32

Consumption rate is 0.08 liters per km: $350 \times 0.08 = 28$ liters.

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